

Electrical Network Analysis (EE2004)

Date: November 2nd, 2024

Course Instructors

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Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 40

Total Questions: 2

Roll No

Section

Student Signature

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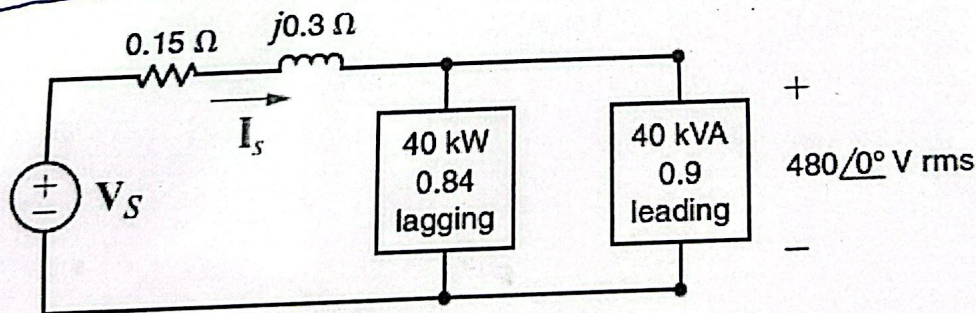
1. Attempt all the questions.
2. Attempt all parts of the same question together.
3. Show all the steps with proper labelled circuit diagrams, and answers with proper units.

CLO #2: Construct power triangle to compute power in AC circuits.

[9 + 3 + 4 + 4 + 5 = 25 marks]

Q1:

The two loads in the circuit shown below can be described as follows:
Load 1 absorbs an average power of 40kW at a lagging power factor of 0.84 and Load 2 absorbs 40kVA at a leading power factor of 0.9.



- a) Construct the power triangle of each load and **compute** the total complex power absorbed by the two loads. Determine the power factor of the combined load.
- b) Compute the impedance of the combined load
- c) Compute the current I_S and the average power loss in the transmission line.
- d) Determine value of source voltage V_S and the power factor of the source.
- e) Given the frequency of the source is 60Hz, compute the value of the capacitor that will correct the power factor to 1 if placed parallel with the two loads.

CLO #3: Analyze balanced three-phase circuits.

Q2:

[3 + 5 + 3 + 4 = 15 marks]

Analyze a balanced three-phase circuit that has the following characteristics:

- Y-Y connected;
 - The **line voltage** at the source, V_{ab} , is $110 \sqrt{3} \angle -60^\circ V_{rms}$;
 - The phase sequence is positive (abc);
 - The line impedance is $3 + j2 \Omega/\phi$;
 - The load impedance is $37 + j28 \Omega/\phi$;
- a) Draw the single phase equivalent circuit diagram for the a-phase.
 - b) Calculate all the line currents.
 - c) Calculate all the phase voltages at the load end.
 - d) Calculate the line voltages at the load end.